

9

(a)

- (i) Distinguish between *bidentate ligand* and *polydentate ligand*.
- (ii) Explain why $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ is paramagnetic and coloured whereas $[\text{Fe}(\text{CN})_6]^{4-}$ is diamagnetic and colourless.
- (iii) Draw the diamagnetic and paramagnetic ions of Fe^{2+} .

[6]

(b)

Describe and explain any one physical property of $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ and $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$.

[4]

- 10 (a) (i) Distinguish between *high spin* and *low spin* complexes.
- (ii) State and explain the observations made when $\text{Cu}_2\text{SO}_{4(s)}$ is dissolved in water and excess ammonia is added to the resultant solution.

(c) The disposal of condensation polymers in landfill sites is more environmentally acceptable than the disposal of addition polymers.

(i) Distinguish between a condensation polymer and an addition polymer.

(ii) Explain why the disposal of condensation polymers in landfill sites is more acceptable.

[4]

[Total: 15]

9 (a) Explain why

- (i) blocks of magnesium are attached to underground iron water pipes,
- (ii) galvanised corrugated iron roofing sheets do not rust even if the zinc coating is scratched,
- (iii) tin food cans quickly rust when the tin coating gets scratched.

[6]

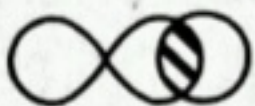
(b) During preparation, phenylamine, is isolated from the mixture by steam distillation.

- (i) Describe how the steam distillation is carried out.
- (ii) Explain why simple distillation is **not** an appropriate method of isolating the components.

4 What is the approximate H – N – H bond angle in $\text{CH}_3\text{C}(=\text{O})\text{NH}_2$?

- A 104.5°
- B 107°
- C 109.5°
- D 120°

5 Which diagram illustrates the bonding in hydrogen chloride?

- A 
- B 
- C 
- D 